

Predicting Falcon 9 First Stage Landings

A SpaceX Launch Data Project

IBM Applied Data Science Capstone · Coursera

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Introduction

SpaceX has reshaped the economics of spaceflight by making its rockets reusable. A Falcon 9 launch costs around \$62 million, while competitors charge upward of \$165 million for a similar launch, largely because SpaceX can recover and reflly the first stage booster instead of building a new one every time. That single difference, whether the first stage lands safely or not, is what makes Falcon 9 launches so much cheaper.

This project sets out to predict, from publicly available launch data, whether a given Falcon 9 first stage will land successfully. Beyond being an interesting prediction problem on its own, this kind of model has a concrete use: if another company is trying to estimate what it would cost to compete with SpaceX for a launch contract, knowing the odds of a successful landing is a meaningful piece of that calculation.

Data Collection

Two independent sources feed into this project. The first is SpaceX's own REST API, which returns structured launch records, but mostly as reference IDs rather than readable details, a launch record points to a rocket ID, a launchpad ID, a payload ID, and so on. Several helper functions query the API again using those IDs to pull back the actual booster name, launch site coordinates, payload mass and orbit, and outcome details for each rocket core, including whether it had landing legs, grid fins, how many times it had flown before, and whether the landing succeeded.

The second source is Wikipedia's running table of every Falcon 9 and Falcon Heavy launch, scraped directly from an archived version of the page using BeautifulSoup so the results stay consistent over time rather than shifting as the page gets updated. Both sources ultimately describe the same set of launches, just with different levels of detail, and get reconciled in the next step.

Data Wrangling

With both data sources in hand, the next job is turning everything into one clean, usable dataset. Falcon 1 launches get filtered out, since the project is specifically about Falcon 9. A handful of records are missing payload mass, those get filled in with the average payload mass across all other launches rather than dropped, to avoid losing otherwise useful rows.

The most important step here is building the actual prediction target, a binary label called **Class** that's 1 if the first stage landed successfully and 0 if it didn't, based on parsing the specific landing outcome

recorded for each launch. Outcomes range from clean successes on a landing pad or drone ship to various flavors of failure or no landing attempt at all.

Exploratory Data Analysis

The data gets explored two ways. First through SQL: the cleaned launch records are loaded into a SQLite database and queried directly, covering things like which launch sites exist, how much total payload mass NASA's CRS missions have carried, the average payload mass for early Falcon 9 versions, when the first successful ground-pad landing happened, and which boosters successfully landed on a drone ship with a mid-sized payload.

Then through visualization: a series of plots look at how launch outcomes relate to flight number, payload mass, launch site, and orbit type, plus a year-by-year view of how the success rate has trended over time. Unsurprisingly, the data shows SpaceX getting visibly better at landings as the program matured, early flights cluster heavily on the failure side, while later flights succeed far more consistently, and that improvement also varies meaningfully by orbit type and payload size.

Geospatial Analysis

Using Folium, an interactive map plots all four Falcon 9 launch sites, with individual launch outcomes marked as color-coded pins clustered at each site, making it easy to see each location's mix of successes and failures at a glance. The map also supports measuring distances directly: by hovering over points of interest near a launch site, like the coastline, a nearby highway, or a railway, and reading off the coordinates, it's possible to calculate how close each site sits to these landmarks, which matters because launch sites tend to be placed near coastlines for safety, in case something goes wrong during ascent.

Interactive Dashboard

The project also includes a small Plotly Dash web application that puts the launch data in front of a user interactively rather than as static charts. A dropdown lets you pick a specific launch site or view all of them together, a pie chart updates to show the split between successful and failed launches for whatever's selected, and a range slider lets you filter by payload mass, with a scatter plot showing how payload mass relates to landing success, colored by booster version, for whatever site and payload range is currently selected.

Predictive Modeling

With a clean dataset and a Class label in hand, the final step is building an actual predictive model. After splitting the data into training and test sets, four classifiers were trained and tuned: Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors, each optimized with 10-fold cross-validated grid search over its own set of hyperparameters.

Model	Cross-Validation Accuracy	Test Accuracy
Logistic Regression	84.6%	83.3%
Support Vector Machine	84.8%	83.3%
Decision Tree	87.5%	83.3%
K-Nearest Neighbors	84.8%	83.3%

On cross-validation, the Decision Tree comes out slightly ahead of the others at 87.5% accuracy, with the rest clustered in the high 84-percent range. Interestingly, all four models land on the exact same accuracy when evaluated on the held-out test set, 83.3%. That's not really a sign that the models are equivalent, it comes down to the test set itself being quite small, just 18 launches, so 83.3% simply means 15 out of 18 correct, a count more than one model can land on even with different decision boundaries underneath.

Conclusion

This project was completed as part of IBM's Applied Data Science Capstone on Coursera, and it works through the full lifecycle of a real data science project: pulling data from an API, scraping a second source from the web, cleaning and combining both, exploring the result through SQL and visualization, mapping it geographically, building an interactive dashboard, and finally training and comparing several classification models.

The end result is a working model that predicts Falcon 9 landing outcomes with reasonable accuracy, and just as importantly, a complete walkthrough of moving from raw, scattered data all the way through to a working predictive model.